PHASE I: BUSINESS PROCESS OVERVIEW

Project Title: Credit Risk Monitoring and Default Warning System (`creditRisk`)

Student Name: Turashimye Kundwa Dorian | **Student ID:** 29540

Database Development with PL/SQL

1. Purpose of the Process

This process covers the **Loan Review, Risk Scoring, and Approval Workflow**. In many small financial institutions, staff review borrower income and loan history manually using spreadsheets. This manual work can cause mistakes, such as approving loans that take up more than half of a borrower's annual income, leading to missed payments and financial loss. The goal of this system is to check every loan request automatically against strict rules before approving money, calculate a risk score for each applicant, and keep a clear record of all decisions.

2. Key Roles and Responsibilities

- **Borrower (Customer):** Submits the loan application with their income, housing status, and job history, then receives the final approval or rejection notice.
- **Loan Officer (Staff):** Verifies the borrower's identity, checks their profile in the system, and enters the new loan request details.
- **Automated System (`creditRisk` System):** Enforces lending rules automatically, calculates the risk score, updates application statuses, and logs any blocked requests.
- **Credit Risk Manager (Supervisor):** Reviews applications with medium or high risk scores, monitors overall risk trends on a reporting dashboard, and makes final decisions on flagged loans.

3. Step-by-Step Process and Decisions

The process begins when a borrower submits an application to a Loan Officer. The officer checks if the borrower already exists in the system (or registers a new profile) and enters details like loan amount, purpose, credit grade, and interest rate.

Once submitted, the system automatically processes the application through three key decision steps:

- **Decision 1 – Income Limit Check:** The system checks if the requested loan payment takes up **50% or less of the borrower's income** and confirms that applicant details are valid. If the loan exceeds 50% of their income, the system

blocks the application immediately, records the rejected attempt, and sends a rejection message to the borrower.

- **Decision 2 – Risk Score Check:** If the loan passes the income limit check, the system saves the pending application and calculates a risk score from 0 to 100 based on debt level, credit rating, and past history. If the **risk score is 40 or lower (Low Risk)**, the system approves the loan automatically.
- **Decision 3 – Manager Review:** If the **risk score is above 40 (Medium or High Risk)**, the application is sent to the Credit Risk Manager. The manager reviews the details and decides to **Approve** or **Reject** the request. The system updates the status and notifies the borrower of the outcome.

- ## 4. System Information Handling

- **RETRIEVE:** The system looks up borrower history and credit rules during initial checks, and pulls up flagged high-risk applications for manager review.
- **CREATE:** New records are saved when adding new borrowers, storing valid loan applications, logging risk evaluations, and recording blocked applications.
- **UPDATE:** The system updates the status of a loan once an automatic approval or manager review is completed.
- **DELETE / ARCHIVE:** Incomplete drafts or canceled applications are removed or archived.

5. Benefits for Daily Operations and Management

For daily operations, this system removes manual income calculations and prevents staff from approving risky loans by mistake. For management, recording risk scores, approval statuses, and blocked attempts allows leaders to use visual dashboards to quickly identify which types of loans carry the highest risk. Ex:kpi’s